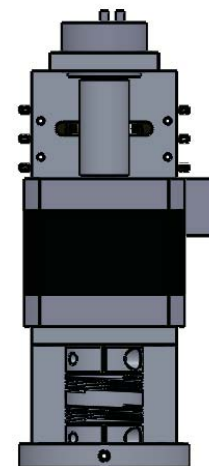
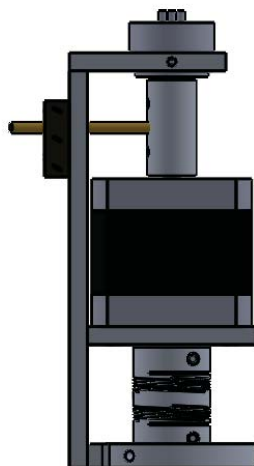
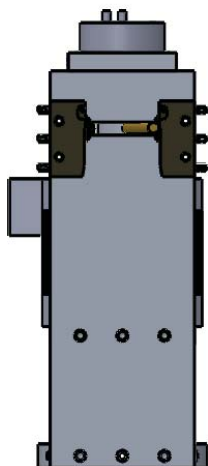
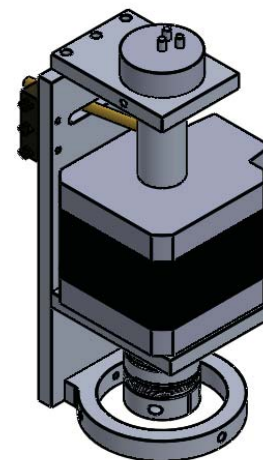
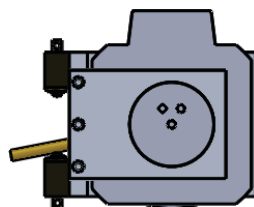
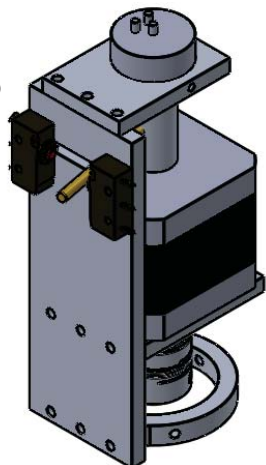


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Napolitano  
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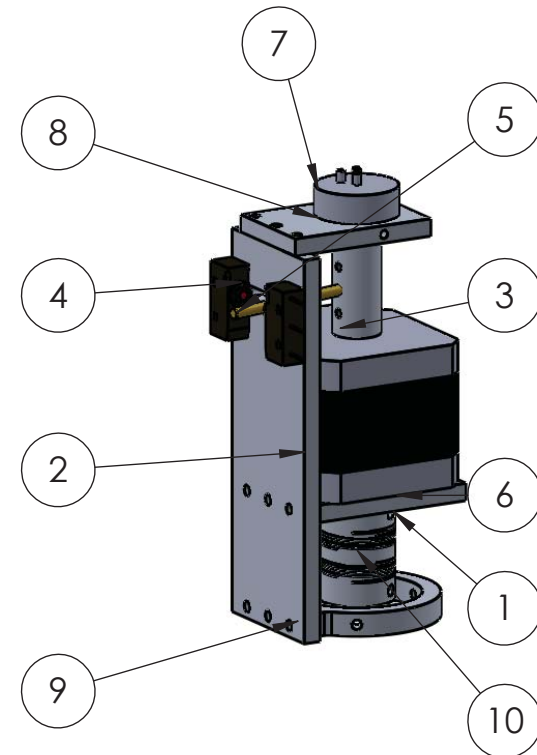


|                                      |  |           |      |                                                                           |                     |
|--------------------------------------|--|-----------|------|---------------------------------------------------------------------------|---------------------|
| UNLESS OTHERWISE SPECIFIED:          |  | NAME      | DATE | <b>TEMPLE UNIVERSITY</b><br>TITLE:<br><h1>Hall A - Moller Target Arm</h1> |                     |
| DIMENSIONS ARE IN INCHES             |  | DRAWN     | JPW  |                                                                           |                     |
| TOLERANCES:                          |  | CHECKED   |      |                                                                           |                     |
| FRACTIONAL $\pm$                     |  | ENG APPR. |      |                                                                           |                     |
| ANGULAR: MACH $\pm$ BEND $\pm$       |  | MFG APPR. |      |                                                                           |                     |
| TWO PLACE DECIMAL $\pm$              |  | Q.A.      |      | SIZE                                                                      | DWG. NO.            |
| THREE PLACE DECIMAL $\pm$            |  | COMMENTS: |      | <b>A</b>                                                                  | ROTATIONAL ELEMENTS |
| INTERPRET GEOMETRIC TOLERANCING PER: |  |           |      | SCALE: 1:2                                                                | REV                 |
| MATERIAL                             |  |           |      |                                                                           |                     |
| FINISH                               |  |           |      |                                                                           |                     |
| DO NOT SCALE DRAWING                 |  |           |      |                                                                           | SHEET 1 OF 5        |

| ITEM NO. | PART NUMBER         | DESCRIPTION                 | QTY. |
|----------|---------------------|-----------------------------|------|
| 1        | Rotary motor mount  | New part                    | 1    |
| 2        | Rotary mount bar    | New part                    | 1    |
| 3        | rot_lim_coup        | New part                    | 1    |
| 4        | microswitch         | Need to order 1?            | 2    |
| 5        | rot_lim_bar         | Need to order               | 1    |
| 6        | Nema-17, dual shaft | Ordered, arrives 10/12/2015 | 1    |
| 7        | Rotary pot          | Using existing              | 1    |
| 8        | rotarty pot mount   | Using existing              | 1    |
| 9        | Actuator_clamp      | Using existing              | 1    |
| 10       | 2464K17             | Using existing              | 1    |

Also need:  
Two 1/4" 4-40 set screws  
Four 3/8" 2-56 screws

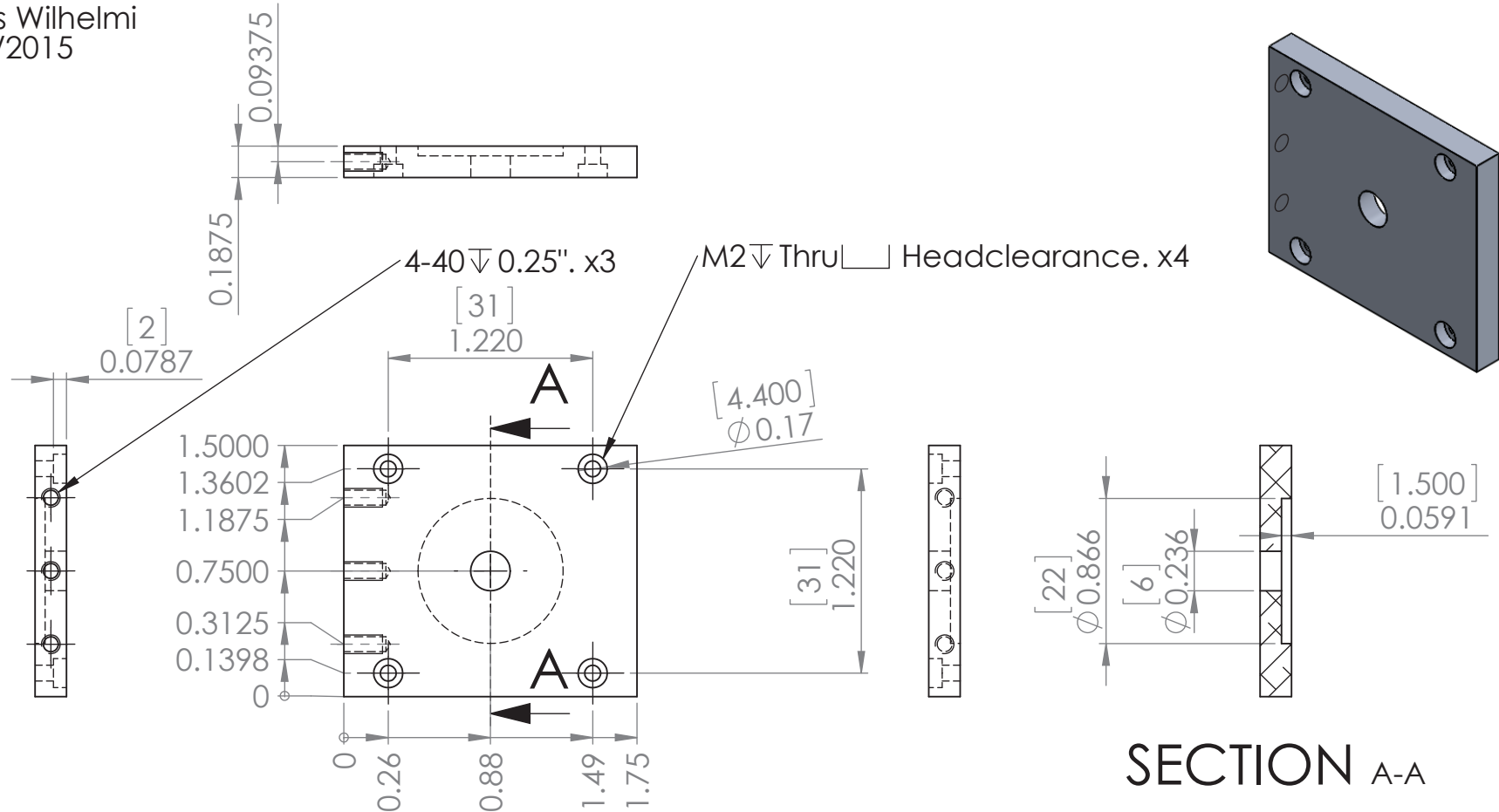
Limits rotation to ~22 degrees total



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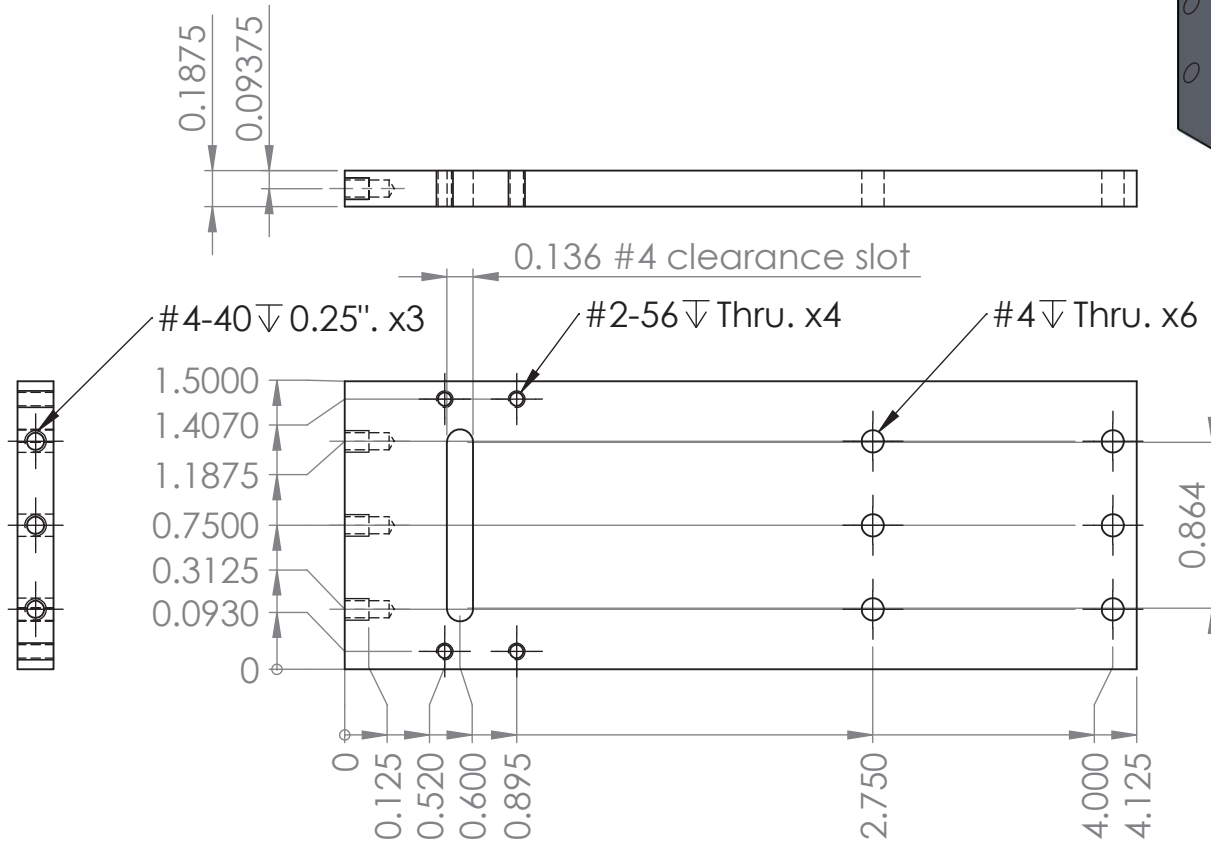
|                                      |  |           |      |                                                                           |                     |          |
|--------------------------------------|--|-----------|------|---------------------------------------------------------------------------|---------------------|----------|
| UNLESS OTHERWISE SPECIFIED:          |  | NAME      | DATE | <b>TEMPLE UNIVERSITY</b><br>TITLE:<br><h2>Hall A - Moller Target Arm</h2> |                     |          |
| DIMENSIONS ARE IN INCHES             |  | DRAWN     | JPW  |                                                                           | 10/07/15            |          |
| TOLERANCES:                          |  | CHECKED   |      |                                                                           |                     |          |
| FRACTIONAL ±                         |  | ENG APPR. |      |                                                                           |                     |          |
| ANGULAR: MACH ± BEND ±               |  | MFG APPR. |      |                                                                           |                     |          |
| TWO PLACE DECIMAL ±                  |  | Q.A.      |      |                                                                           |                     |          |
| THREE PLACE DECIMAL ±                |  | COMMENTS: |      |                                                                           |                     |          |
| INTERPRET GEOMETRIC TOLERANCING PER: |  |           |      | SIZE                                                                      | DWG. NO.            | REV      |
| MATERIAL                             |  |           |      | <b>A</b>                                                                  | Rotational elements | <b>2</b> |
| FINISH                               |  |           |      | SCALE: 1:2                                                                |                     | WEIGHT:  |
| DO NOT SCALE DRAWING                 |  |           |      | SHEET 2 OF 5                                                              |                     |          |

Item 1 on Rotational elements bill of materials.  
James Wilhelm  
10/07/2015



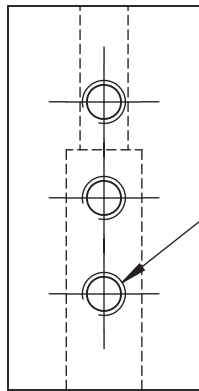
|                                             |  |            |      |                                                                          |                                                                  |  |              |  |
|---------------------------------------------|--|------------|------|--------------------------------------------------------------------------|------------------------------------------------------------------|--|--------------|--|
| UNLESS OTHERWISE SPECIFIED:                 |  | NAME       | DATE | TEMPLE UNIVERSITY<br>TITLE:<br><div>Hall A - Moller<br/>Target Arm</div> |                                                                  |  |              |  |
| DIMENSIONS IN INCHES<br>TOLERANCES: ±0.0005 |  | DRAWN      | JPW  |                                                                          |                                                                  |  | 10/07/15     |  |
|                                             |  | CHECKED    |      |                                                                          |                                                                  |  |              |  |
|                                             |  | ENG APPR.  |      |                                                                          |                                                                  |  |              |  |
|                                             |  | MFG APPR.  |      |                                                                          |                                                                  |  |              |  |
| INTERPRET GEOMETRIC<br>TOLERANCING PER:     |  | Q.A.       |      |                                                                          | SIZE<br><b>A</b> DWG. NO.<br>Rotational elements REV<br><b>2</b> |  |              |  |
| MATERIAL<br>Aluminum                        |  | COMMENTS:  |      |                                                                          |                                                                  |  |              |  |
| FINISH                                      |  |            |      |                                                                          |                                                                  |  |              |  |
| DO NOT SCALE DRAWING                        |  | SCALE: 1:2 |      |                                                                          | WEIGHT:                                                          |  | SHEET 3 OF 5 |  |

Item 2 in Rotational elements bill of materials.  
James Wilhelmi  
10/07/2015

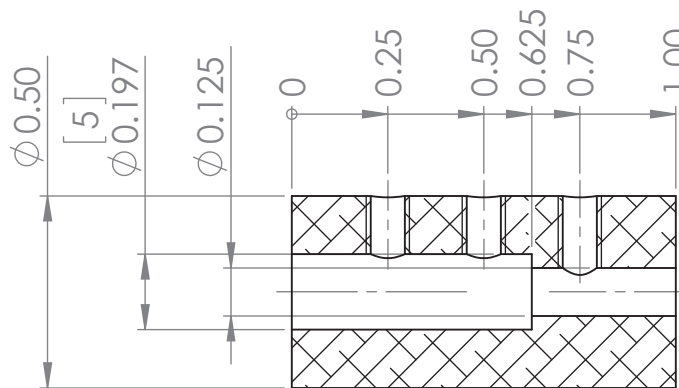
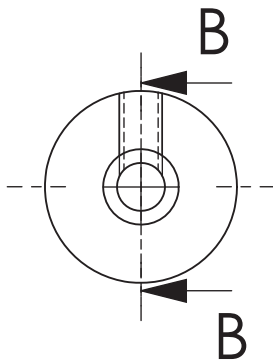


|                                             |  |           |      |                                                                          |                                                                              |         |              |
|---------------------------------------------|--|-----------|------|--------------------------------------------------------------------------|------------------------------------------------------------------------------|---------|--------------|
| UNLESS OTHERWISE SPECIFIED:                 |  | NAME      | DATE | TEMPLE UNIVERSITY<br>TITLE:<br><div>Hall A - Moller<br/>Target Arm</div> |                                                                              |         |              |
| DIMENSIONS IN INCHES<br>TOLERANCES: ±0.0005 |  | DRAWN     | JPW  |                                                                          |                                                                              |         | 10/07/15     |
|                                             |  | CHECKED   |      |                                                                          |                                                                              |         |              |
|                                             |  | ENG APPR. |      |                                                                          |                                                                              |         |              |
|                                             |  | MFG APPR. |      |                                                                          |                                                                              |         |              |
| INTERPRET GEOMETRIC<br>TOLERANCING PER:     |  | Q.A.      |      |                                                                          | SIZE<br><b>A</b> <div>DWG. NO.<br/>Rotational elements</div> REV<br><b>2</b> |         |              |
| MATERIAL<br>Aluminum                        |  | COMMENTS: |      |                                                                          |                                                                              |         |              |
| FINISH                                      |  |           |      |                                                                          |                                                                              |         |              |
| DO NOT SCALE DRAWING                        |  |           |      |                                                                          | SCALE: 1:2                                                                   | WEIGHT: | SHEET 4 OF 5 |

Item 3 in Rotational elements bill of materials  
 James Wilhelmi  
 10/07/2015



#4-40 to center.



SECTION B-B

SCALE 2 : 1

|                                             |  |           |      |                                                                          |                  |  |  |                                 |  |                 |
|---------------------------------------------|--|-----------|------|--------------------------------------------------------------------------|------------------|--|--|---------------------------------|--|-----------------|
| UNLESS OTHERWISE SPECIFIED:                 |  | NAME      | DATE | TEMPLE UNIVERSITY<br>TITLE:<br><div>Hall A - Moller<br/>Target Arm</div> |                  |  |  |                                 |  |                 |
| DIMENSIONS IN INCHES<br>TOLERANCES: ±0.0005 |  | DRAWN     | JPW  |                                                                          |                  |  |  | 10/07/15                        |  |                 |
|                                             |  | CHECKED   |      |                                                                          |                  |  |  |                                 |  |                 |
|                                             |  | ENG APPR. |      |                                                                          |                  |  |  |                                 |  |                 |
|                                             |  | MFG APPR. |      |                                                                          |                  |  |  |                                 |  |                 |
| INTERPRET GEOMETRIC<br>TOLERANCING PER:     |  | Q.A.      |      |                                                                          | SIZE<br><b>A</b> |  |  | DWG. NO.<br>Rotational elements |  | REV<br><b>2</b> |
| MATERIAL<br>Aluminum                        |  | COMMENTS: |      |                                                                          |                  |  |  |                                 |  |                 |
| FINISH                                      |  |           |      |                                                                          |                  |  |  |                                 |  |                 |
| DO NOT SCALE DRAWING                        |  |           |      |                                                                          |                  |  |  |                                 |  |                 |

REV  
**2**